WEEK 1 – DESIGN DOCUMENT

PS-062

Diagnostic Lab Test Ordering and Report TAT Engine

Domain: Healthcare – Diagnostics
Project ID: T083-PS-062
Technology: Java 17+ or Python 3.10+
Project Type: Command-file-driven Backend System

Team Members: _____

Institution / Class: _____

Submission Date: _____

1. Project Overview

The Diagnostic Lab Test Ordering and Report TAT Engine is a backend system designed to automate the operations of a diagnostic laboratory chain. The system accepts patient test orders, validates sample requirements and fasting conditions, assigns collection slots, routes samples to suitable laboratories, calculates promised turnaround time (TAT), tracks sample status, calculates billing and generates operational reports.

The system also handles sample rejection, recollection, critical results, delayed reports, cancellations and home collection. It is command-file-driven and does not require a graphical user interface.

2. Project Objectives

- Register diagnostic tests, panels, laboratories and phlebotomists.
- Validate patient orders based on fasting requirements, sample types, tube requirements and collection availability.
- Route samples to appropriate laboratories based on tier, test capability and daily capacity.
- Calculate promised and actual turnaround times using priority, courier delays and laboratory operating hours.
- Track sample lifecycle events, including collection, processing, verification, rejection and recollection.
- Calculate order prices using panel pricing, priority surcharges, home-collection fees and delayed-report credits.
- Generate order status, TAT breach and daily laboratory capacity reports.

3. System Scope

3.1 In Scope

- Test master registration
- Panel registration
- Laboratory registration
- Phlebotomist registration and slot management
- Patient order creation and validation
- Fasting validation and waiver
- Tube-sharing calculation
- Laboratory routing
- TAT calculation
- Sample lifecycle tracking
- Sample rejection and recollection
- Critical value verification
- Pricing and delayed-report credits
- Cancellation and report release management
- Daily capacity and TAT breach reports

3.2 Out of Scope

- Graphical user interface
- Online payment gateway integration
- Direct hospital information system integration
- Real-time SMS and email notifications
- Actual laboratory analyser hardware integration
- Electronic medical record management

4. Glossary

Term	Description
TAT	Turnaround Time: time taken from sample collection to report delivery.
Diagnostic Test	A medical test performed on a patient sample.
Test Master	Collection of registered tests and their requirements.
Panel	A predefined group of diagnostic tests sold at a bundled price.
Sample	Biological material collected from a patient.
Sample Type	BLOOD, URINE, SWAB or TISSUE.
Tube Type	Container required to collect a particular sample.
Fasting	Required period during which a patient must not consume food before collection.
Priority	ROUTINE, URGENT or STAT order classification.
Lab Tier	Laboratory classification: L1, L2 or REFERENCE.
Routing	Assignment of a test sample to an appropriate laboratory.
Courier Cut-off	Latest collection time for same-day dispatch to a higher-tier laboratory.
Phlebotomist	Healthcare professional responsible for collecting patient samples.
Recollection	Collection of a replacement sample after rejection.
Critical Value	Result outside the predefined critical range requiring doctor verification.
Panel Pricing	Bundled price for a predefined group of tests.
Delayed-report Credit	Credit applied when a report exceeds its promised delivery time by the specified threshold.
Report Release	Making verified test results available to the patient.
Capacity	Maximum number of samples a laboratory can process per day for each sample type.
NON_FASTING	Flag assigned to affected results when fasting requirements are waived.

5. System Architecture

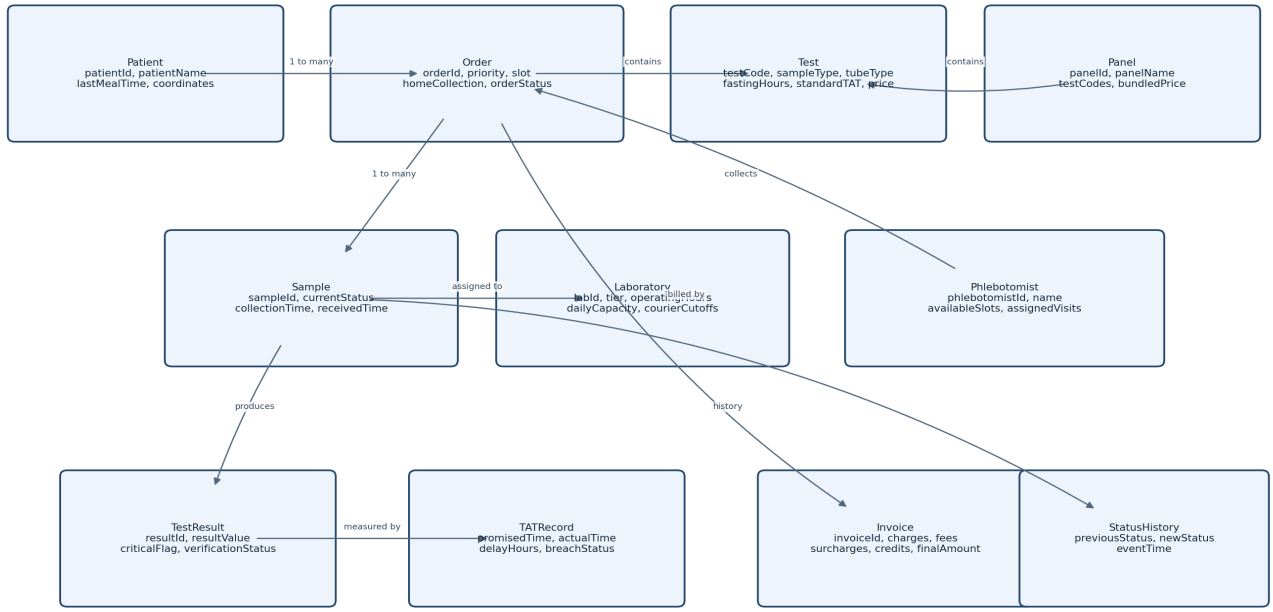
The system follows a modular service-based architecture. Each module is responsible for a specific business function and communicates through clearly defined data models and service operations.

Module	Responsibility
Master Data Module	Registers tests, panels, laboratories and phlebotomists.
Order Management Module	Creates and validates patient orders.
Fasting and Sample Validation Module	Checks fasting conditions, sample requirements and tube sharing.
Collection Management Module	Assigns home-collection slots and phlebotomists.
Lab Routing Module	Selects laboratories based on tier, test availability and capacity.
TAT Engine	Calculates promised and actual report times.
Sample Lifecycle Module	Tracks sample status and handles rejection and recollection.
Pricing Module	Calculates test charges, panel charges, surcharges and credits.
Report Management Module	Handles doctor verification and patient report release.
Reporting Module	Generates daily capacity, order status and TAT breach reports.

Overall flow: Patient Order → Validation → Collection → Lab Routing → Sample Processing → Result Generation → Verification → Report Release → Billing and Reporting.

6. Class Diagram

PS-062 – Main Domain Class Diagram



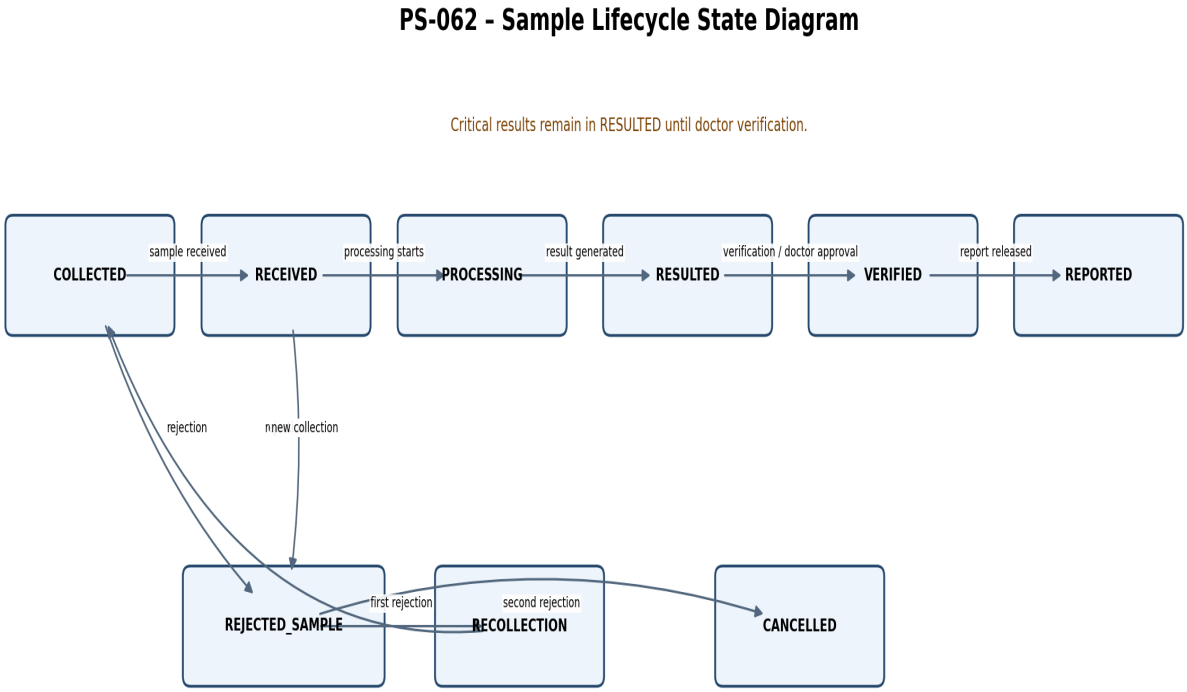
The following diagram shows the main domain classes and their relationships. The detailed attributes are listed below the diagram.

Class	Important attributes	Purpose
Patient	patientId, patientName, lastMealTime, latitude, longitude	Represents the person placing diagnostic orders.
Order	orderId, patientId, priority, homeCollection, collectionSlot, orderStatus, totalAmount	Represents one patient order containing tests and/or panels.
Test	testCode, sampleType, tubeType, fastingHours, standardTATHours, labTier, price, criticalRange	Defines a diagnostic test and its requirements.
Panel	panelId, panelName, testCodes, bundledPrice	Defines a group of tests sold at a bundled price.
Laboratory	labId, labName, tier, operatingHours, dailyCapacity, courierCutoffTimes, supportedTests	Represents a processing laboratory.
Sample	sampleId, orderId, sampleType, tubeType, currentStatus, collectionTime, receivedTime	Tracks a collected sample and its lifecycle.
Phlebotomist	phlebotomistId, name, availableSlots, assignedVisits	Represents a sample-collection professional.
TestResult	resultId, testCode, resultValue, criticalFlag, verificationStatus, reportTime	Stores a test result and its verification state.
TATRecord	promisedTime, actualTime, delayHours, breachStatus	Stores promised and actual TAT information.
Invoice	invoiceId, orderId, testCharges, panelCharges, collectionFee, prioritySurcharge, credits, finalAmount	Stores the order's billing calculation.
StatusHistory	historyId, sampleId, previousStatus, newStatus, eventTime	Stores each sample status transition.

6.1 Class Relationships

- One patient can create multiple orders.
- One order can contain multiple tests and panels.
- One panel contains multiple tests.
- Multiple tests may share a sample tube when sample type and tube type match, subject to the four-tests-per-tube limit.
- One laboratory can process multiple samples.
- One phlebotomist can handle multiple collection visits within the daily limits.
- One sample can have multiple status-history records.
- Each test execution has an associated result and TAT record.
- One order has an associated invoice.

7. Sample State Diagram



A sample normally moves through COLLECTED → RECEIVED → PROCESSING → RESULTED → VERIFIED → REPORTED. Rejection and recollection are exception branches.

Current State	Event / Condition	Next State
COLLECTED	Sample received by the laboratory	RECEIVED
RECEIVED	Processing begins	PROCESSING
PROCESSING	Result generated	RESULTED
RESULTED	Normal verification completed	VERIFIED
RESULTED	Critical result awaits doctor verification	Remains RESULTED until doctor command
VERIFIED	Report released according to release rules	REPORTED
COLLECTED or RECEIVED	Sample marked haemolysed, clotted or insufficient	REJECTED_SAMPLE

Current State	Event / Condition	Next State
REJECTED_SAMPLE	First rejection; recollection scheduled	RECOLLECTION
RECOLLECTION	New sample collected; TAT restarts	COLLECTED
REJECTED_SAMPLE	Second rejection on same order for affected test	CANCELLED

Design note: Order status and individual test/sample statuses must be maintained separately because different tests in one order may progress at different speeds.

8. Business Rule Register

Each business rule is assigned a unique identifier so that it can be traced to implementation logic and test cases.

Rule ID	Rule Name	Business Rule / Expected Behaviour
BR-01	Test Registration	Each test must have a unique code and include sample type, tube type, fasting hours, standard TAT, lab tier and price.
BR-02	Panel Registration	A panel must contain valid registered tests and a bundled price.
BR-03	Laboratory Registration	Each lab must have a valid tier, operating hours, daily sample-type capacity and courier cut-off information.
BR-04	Phlebotomist Registration	Each phlebotomist must have available collection slots. Each slot is 30 minutes.
BR-05	Fasting Validation	The longest fasting requirement among ordered tests governs. Collection is valid when collection time minus last-meal time is greater than or equal to required fasting hours. Otherwise the order becomes NEEDS_RESCHEDULE unless waived.
BR-06	Fasting Waiver	If the patient waives fasting, the order may proceed and affected test results must be flagged NON_FASTING.
BR-07	Tube Sharing	Tests with the same sample type and tube type may share one tube. A tube serves at most four tests; a fifth test requires another tube.
BR-08	Order Validation	An order must contain valid tests or panels, a valid priority and a valid collection slot before acceptance.
BR-09	Phlebotomist Slot Duration	Each collection slot must be 30 minutes.
BR-10	Daily Visit Limit	A phlebotomist can perform at most eight home-collection visits per day.
BR-11	Travel Distance	Straight-line distance between consecutive assigned home visits must not exceed 10 km.
BR-12	Home Collection Distance	Home collection is available up to 25 km: Rs 200 at or below 10 km; Rs 400 above 10 km and up to 25 km; unavailable beyond 25 km.
BR-13	Home Collection Fee Waiver	Home-collection fee is waived when order value is greater than or equal to Rs 2,000.
BR-14	Lowest-Tier Routing	Assign a test to the lowest-tier lab capable of performing it and with remaining daily capacity for the required sample type.
BR-15	Higher-Tier Routing	If a lab cannot perform the test or has no capacity, route to the next eligible higher tier.
BR-16	Courier Cut-off	Collection after cut-off dispatches the next day. Collection exactly at cut-off is eligible for same-day dispatch.
BR-17	Reference Lab TAT	Add 24 hours of courier time for REFERENCE-tier tests.
BR-18	Priority Multipliers	ROUTINE = 1.0x; URGENT = 0.5x; STAT = 0.25x.
BR-19	Minimum TAT	Calculated processing TAT must be at least two hours.
BR-20	TAT Clock Start	TAT clock starts at COLLECTED.
BR-21	Promised Report Time	Calculate TAT using standard test TAT, priority multiplier, minimum two-hour floor and applicable courier time. Round the promised report time up to the next hour.
BR-22	Operating Hours	Processing occurs only within lab operating hours. If received outside hours, RECEIVED time is the next opening time. TAT clock continues continuously.
BR-23	STAT Availability	STAT samples may be accepted 24x7 at L1 labs only.
BR-24	Sample Rejection	Reject samples marked haemolysed, clotted or insufficient.
BR-25	Free Recollection	Schedule recollection free of charge within 24 hours after a rejection.
BR-26	TAT Restart	After recollection, restart the TAT clock from the new collection time.
BR-27	Second Rejection	A second rejection cancels affected tests and triggers a full refund for those tests.
BR-28	Critical Value Detection	Mark a result critical when it is outside the supplied critical range.
BR-29	Doctor Verification	A critical result cannot move to VERIFIED until a doctor-verification command is received.

Rule ID	Rule Name	Business Rule / Expected Behaviour
BR-30	Critical Report Deadline	A critical report must be REPORTED within one hour after VERIFIED; otherwise record a critical breach regardless of normal promised TAT.
BR-31	Panel Pricing	Apply the bundled panel price when all member tests of the panel are ordered.
BR-32	Duplicate Test Removal	If an individual test is also included in an ordered panel, execute it once and do not charge it separately.
BR-33	Overlapping Panels	Charge both overlapping panels at their bundled prices, but execute shared tests only once.
BR-34	Priority Surcharge	URGENT adds 50%; STAT adds 100%; ROUTINE adds no surcharge. Apply to the relevant test or panel price.
BR-35	Tube Collection Fee	Charge Rs 50 for each tube beyond the second tube.
BR-36	Delayed Report Credit	If actual report time exceeds promised time by more than two hours, credit 10% of affected test price. If delay exceeds 24 hours, credit 50%.
BR-37	Credit Restrictions	Do not issue delayed-report credits when delay is caused by patient-side recollection or a waived fasting breach.
BR-38	Credit Limit	Total credits for a test cannot exceed that test's price.
BR-39	Pre-Collection Cancellation	Cancellation before COLLECTED is free.
BR-40	Post-Collection Cancellation	Cancellation after COLLECTED but before PROCESSING incurs a charge of 50% of the test price.
BR-41	Processing Cancellation	No refund is allowed after PROCESSING begins.
BR-42	Panel Cancellation	Panel cancellation must be all-or-nothing.
BR-43	Normal Report Release	Release the complete order report only after every test in the order reaches VERIFIED, subject to stated exceptions.
BR-44	STAT Report Release	A STAT test report may be released individually after verification without waiting for remaining tests.
BR-45	Partial Report Release	Release a partial report when a test is cancelled due to a second sample rejection.
BR-46	Order Status Report	Print current order status and individual test statuses.
BR-47	TAT Breach Report	Identify tests breaching promised TAT and critical-report deadlines.
BR-48	Daily Capacity Report	For any requested date, show lab capacity used and remaining by sample type.

9. Edge Cases and Expected Behaviour

Edge Case	Expected Behaviour
Collection exactly N hours after last meal	Fasting validation passes because the condition is greater than or equal to N hours.
Fifth test needing same tube type	A second tube is created; no tube contains more than four tests.
Collection exactly at courier cut-off	Same-day courier dispatch is allowed.
STAT test with 4-hour standard TAT	0.25 × 4 = 1 hour; the two-hour minimum applies.
Sample received one minute before lab closes	It may be received before closing; processing is constrained by lab operating hours.
Panel + member test + second overlapping panel	Charge both panel prices; execute shared tests only once.
Home collection exactly 10 km away	Rs 200 fee applies unless the order-value waiver applies.
Order value exactly Rs 2,000	Home-collection fee is waived.
Report delayed exactly 2 hours	No delayed-report credit because the rule says more than two hours.
Second rejection	Affected tests are cancelled and fully refunded.

10. Key Design Decisions and Assumptions

- Track individual tests independently even when they belong to the same order.
- Use consistent date-time values and a documented timezone for all input and output files.
- Calculate TAT separately for each test because tests may have different priorities, routes and standard TAT values.
- Laboratory operating hours affect processing availability but do not stop the TAT clock.

- Separate pricing logic from test execution logic so overlapping panels can be charged correctly while shared tests run once.
- Store every sample lifecycle transition in status history.
- Track critical breaches separately from ordinary TAT breaches.
- Use deterministic processing so identical input files produce identical output files.

11. Clarifications to Confirm with the Coordinator

- How exactly should operating hours and courier waiting time be incorporated into the promised report timestamp, beyond the explicit 24-hour REFERENCE-tier addition?
- Does a test's lab tier indicate its minimum required tier, or should routing use a separate list of labs capable of performing that test?
- How should a home-collection visit be assigned to a lab when more than one lab is eligible?
- Does a second rejection cancel only the affected test or all tests sharing the rejected physical tube? The brief says affected tests; the implementation should record the exact affected set.
- How should order value for the Rs 2,000 home-collection waiver be calculated when panels, surcharges or discounts are present?
- How should a panel cancellation interact with the 50% post-collection charge and the no-refund-after-processing rule?

These questions do not prevent the design from proceeding. The team's chosen interpretations should be recorded in the decision log before implementation.

12. Conclusion

The proposed Diagnostic Lab Test Ordering and Report TAT Engine provides an automated approach to managing diagnostic laboratory operations. It combines order validation, sample management, laboratory routing, TAT calculation, pricing and report release in one backend system. The modular design allows business rules to be implemented and tested independently and supports future enhancements such as delta-checking and batch-based analyser scheduling.